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PERSPECTIVE

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Detecting and interpreting higher-order interactions in ecological communities

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Abstract

When species simultaneously compete with two or more species of competitor, higher-order interactions (HOIs) can lead to emergent properties not present when species interact in isolated pairs. To extend ecological theory to multi-competitor communities, ecologists must confront the challenges of measuring and interpreting HOIs in models of competition fit to data from nature. Such efforts are hindered by the fact that different studies use different definitions, and these definitions have unclear relationships to one another. Here, we propose a distinction between 'soft' HOIs, which identify possible interaction modification by competitors, and 'hard' HOIs, which identify interactions uniquely emerging in systems with three or more competitors. We show how these two classes of HOI differ in their motivation and interpretation, as well as the tests one uses to identify them in models fit to data. We then show how to operationalise this structure of definitions by analysing the results of a simulated competition experiment underlain by a consumer resource model. In the course of doing so, we clarify the challenges of interpreting HOIs in nature, and suggest a more precise framing of this research endeavour to catalyse further investigations.

KEYWORDS
annual plants, interaction modification, model fitting

INTRODUCTION

A central paradox in the study of species coexistence is that while ecologists aim to understand the maintenance of diverse assemblages of species in nature, efforts to achieve this goal typically focus on the interactions between pairs of species. One consequence of this approach is that we cannot rule out the possibility that the impressive diversity we observe depends on mechanisms of coexistence that only emerge in systems of more than two species (Billick & Case, 1994; Levine et al., 2017).

Particularly vexing is the possible operation of higher-order interactions (HOIs), where competitors modify the interactions between other competitors in the system. If these interactions are significant, even a perfect understanding of the interaction between each and every pair of species in isolation would poorly describe the dynamics of all species simultaneously interacting (Billick & Case, 1994; Levine et al., 2017; Neill, 1974). Thus, among the most fundamental questions for community ecologists is whether we can build an adequate theoretical understanding of species coexistence while ignoring HOIs. In principle, it is quite possible that our current understanding is incomplete. For example, with few exceptions, the mutual invasibility criterion can predict the coexistence of species pairs, and as a consequence, almost all of modern coexistence theory is built on this criterion (Chesson, 2000; Grainger et al., 2019; Levine et al., 2017). However, when species modify the interactions between one another, three species can coexist even when all pairs of competitors fail to meet this criterion (Grilli et al., 2017). Observations such as this could in principle explain the general failure of empirical studies focused on pairwise interactions to predict the coexistence of many species (e.g. Kraft et al., 2015).

Despite the centrality of the HOI question for community ecology, progress towards its resolution has been

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slowed by confusion over how to precisely define these interactions and therefore identify what ecologists are searching for. As a consequence, different studies treat different interactions and processes as HOIs (e.g. compare Letten & Stouffer, 2019 to Adler & Morris, 1994), and it is not clear how the different uses of the term relate to one another (Sanchez, 2019). The elusiveness of a precise mathematical definition may seem surprising given that most ecologists would agree to a verbal definition of HOIs involving interactions where one species modifies the interaction between at least another two. Indeed, 'interaction modification' has sometimes been used interchangeably with HOI. Nonetheless, it has been challenging to agree on exactly what constitutes an interaction modification (Adler & Morris, 1994; Billick & Case, 1994; Letten & Stouffer, 2019; Pomerantz, 1981; Sanchez, 2019). It is further unclear whether interaction modification is synonymous with the processes that uniquely complicate our conceptualisation of coexistence in diverse communities, one of the major motivations for studying HOIs in the first place. Here we pose that defining HOIs based on their unique emergence in systems of more than two species might be a more fruitful path forward than using math to precisely identify interaction modification.

There are several reasons why the traditional approach to defining HOIs based solely on interaction modification is challenging. First, as we will explain in the first half of this article, such definitions tend to include effects that are perfectly resolvable in pairwise interaction studies – namely nonlinear density dependence (Adler & Morris, 1994) – violating one major motivation for this research endeavour. To make matters worse, efforts to identify HOIs almost uniformly rely on phenomenological models of competition and interpretation of the included parameters. Without a mechanistic understanding of the underlying processes involved, it can be challenging to agree on what is or is not interaction modification, a point we elaborate on in the next section. Further complications arise because previous definitions of HOIs were developed with a small range of classical competition models in mind. Since that time, new statistical modelling software allows ecologists to fit a much wider range of interaction models (Mayfield & Stouffer, 2017). This increase in model flexibility requires a general definition for HOIs that can be applied to any density-dependent model of population dynamics. Finally, it is helpful to acknowledge that ecologists have different motivations for studying HOIs, and these varying motivations might aim for different elements of a definition. For example, a definition for HOIs that is motivated by improving the statistical fit of a model to interaction data may not be sufficient if the goal is to test whether *n*-species dynamics can be predicted from empirical measurements of the dynamics of all species pairs (Friedman et al., 2017).

In this article, we propose a nested structure of operational definitions for HOIs that accounts for the myriad of density dependence forms possibly observable in natural communities. We first summarise the simple Lotka Volterra case and the limitations of past mathematical attempts to define HOIs. We then propose a distinction between 'soft' HOIs that include interaction modification broadly defined, and 'hard' HOIs that specifically aim to identify interaction modifications unknowable from studies of pairwise interactions. Critically, depending on how HOIs are defined, and particularly in relation to nonlinear density dependence, different empirical and statistical approaches are required to quantify their role. Thus, in the second part of the article, we illustrate the experimental designs and analyses required to operationalise the proposed structure of definitions. We do so by fitting phenomenological models to data derived from simulating a virtual competition experiment where the mechanistic processes determining competition are fully known (Letten & Stouffer, 2019). An alternative approach involves directly evaluating interaction modification through empirical observations or experiments (Adler & Morris, 1994), as is common in the microbial literature (Kelsic et al., 2015; Mickadlide & Kuehn, 2019). However, even this approach would benefit from quantitative definitions for HOIs of the sort we develop here (Sanchez, 2019).

Primer on higher-order interactions in Lotka Volterra models

For the purpose of defining HOIs, we focus on modelling a focal species' performance (usually per capita population growth rate) as a function of the population density of multiple species of competitor. This can be expressed generally as,

$$\frac{1}{n_i}\frac{dn_i}{dt} = F_i\left(n_1, \ldots n_m\right), \tag{1}$$

where F_i gives the per capita population growth rate of the focal species i, and $n_1, \ldots n_m$ are the population densities of competitor species one through m in the community, including the population density of the focal species, n_i. An analogous equation holds for population growth rate over discrete time intervals: $\frac{n_i(t+1)}{n_i(t)} = F_i\left(n_1, \ldots n_m\right)$. In most widely used models of species interactions, each competitor has one effect on itself and one effect on each of the other species in the community. The simplest example of such a pairwise competition model is the Lotka-Volterra (LV) model:

$$\frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \sum_{j=1}^{m}\alpha_{ij}n_j\right), \tag{2}$$

where r_i is the intrinsic rate of growth of the focal species i, and α_{ij} is the per capita effect of competitor j on the growth

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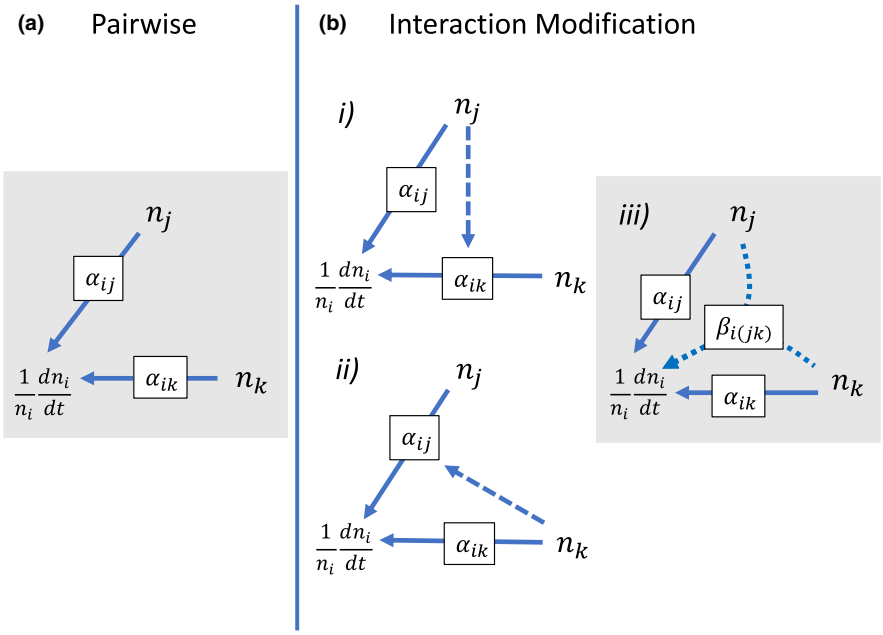


FIGURE 1 How interaction modifications lead to higher-order interactions (HOIs). In A, a pairwise model is shown without interaction modification. The competitive effects of species *j* and *k* on the per capita growth of the focal species *i*, are shown as separate blue arrows. These effects may be simple per capita competition coefficients, α_{ij} and α_{ik}, or could be more complicated non-linear functions of the density of a single species. In B, a model with interaction modification is shown. In (i) the dashed arrow shows that the effect of species *k* is modified by the density of species *j*, and in (ii) the effect of *j* is modified by the density of *k*. In reality, one cannot assign either species as the modifier; rather they modify each other's effects in a way that emerges as a single HOI. The HOI in this case is quantified by introducing the new parameter $\beta_{i(jk)}$, and shown with the curved arrows in (iii)

rate of the focal species. This model is pairwise because each interaction is specified by the pair of species involved-the focal species *i* and the competitor *j*. The defining property of any pairwise model, such as the Lotka Volterra model, is that the per capita effect of each species of competitor is independent of the densities of any *other* species of competitor (Figure 1a).

By contrast, HOIs are commonly thought to disrupt pairwise competition. Interaction modification occurs when the effect of one competitor species is modified by the density of another competitor species (Adler & Morris, 1994). We can introduce an interaction modification into the Lotka Volterra model by replacing any of the constant terms α_{ij} with a function of the density of another competitor (Billick & Case, 1994). For instance, in the following Lotka Volterra model, the focal species *i*'s performance is dependent on two competitor species, *j* and *k* (intraspecific competition is ignored for simplicity):

$$\frac{1}{n_i}\frac{dn_i}{dt} = F_i(n_j, n_k) = r_i(1 - \alpha_{ij}n_j - \alpha_{ik}n_k). \quad (3)$$

Replacing the term α_{ij} with the expression $\alpha_{ij} + \beta_{i(jk)}n_k^x$, makes the per capita effect of species *j* dependent on the density of another competitor, n_k. More specifically the parameter $\beta_{i(jk)}$ measures the strength of this interaction modification (Figure 1b). The power *x* defines the non-linearity of the relationship between interaction modification and the density of species *k*. Substituting this function into the model in Equation 3 introduces the product of competitors *j* and *k* (the latter raised to the power *x*) as an additional term:

$$\frac{1}{n_i}\frac{dn_i}{dt} = F_i(n_j, n_k) = r_i(1 - \alpha_{ij}n_j - \alpha_{ik}n_k - \beta_{i(jk)}n_j n_k^x). \quad (4)$$

HOIs such as these imply that competition is functionally different when more than one competitor species is present and that there are emergent properties in the community that cannot be predicted by pairwise effects. Specifically, the interaction strength $\beta_{i(jk)}$ is unknowable from pairwise interaction experiments. Note that if n_j or n_k equals zero, the models in Equations 3 and 4 are identical. HOIs such as $\beta_{i(jk)}$ may suggest specific biological hypotheses: something about the behaviour or traits of the competitors are functionally distinct when they are together compared with when they are separate. Importantly, an interaction modification cannot be attributed to any one competitor – rather it is an emergent property of the multi-species system (Figure 1b). Nonetheless, the contribution of the competitors to the HOI need not be symmetric, as occurs when $x \neq 1$ in Equation 4.

A seemingly straightforward mathematical definition for HOIs in the Lotka Volterra context can be obtained by taking the partial derivative of the growth function, F_i in Equation 1, with respect to the density of a single competitor species *j*, $\frac{\partial F_i}{\partial n_j}$. In general, and as explained in calculus textbooks, a partial derivative quantifies how a

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function of multiple variables (in this context- the growth function F_i) changes with an infinitesimally small change in one of those variables (in this context- the density of species j, n_j). $\frac{\partial F_i}{\partial n_j}$ defines the focal species i's growth sensitivity to the density of species j. If this partial derivative is a function of the density of any species other than i and j, then it could be argued that there are HOIs: mathematically, the density of one species is modifying the effect of another. This is similar to definitions used in earlier discussions of HOIs based on Lotka Volterra competition (Case & Bender, 1981), and closely follows the verbal argument that HOIs emerge when the effect of one competitor on another depends on any other competitor. For the Lotka Volterra model without $\beta_{i(jk)}$ (Equation 3) the partial derivative, $\frac{\partial F_i}{\partial n_j}$ is simply $-r_i\alpha_{ij}$, while for the case with $\beta_{i(jk)}$ (Equation 4), we get $-r_i\alpha_{ij} - r_i\beta_{i(jk)}n_k^x$. Only in the latter case does the effect of species j on i depend on the density of species k, and a HOI, therefore, identified. In sum, in the Lotka Volterra case, a partial derivative definition for a HOI is straightforward, and the interaction modification it identifies as a HOI is unknowable from pairwise interactions alone. Unfortunately, defining HOIs becomes far more complicated against a backdrop of nonlinear effects of competition on growth.

The challenge of identifying interaction modification with nonlinear relationships between growth and competition

Although the prior section captures the essential connection between interaction modification and HOIs, ecologists lack a general definition for HOIs that specifically identifies the interaction modifications that only emerge in systems with more than pairwise interactions (Billick & Case, 1994; Grilli et al., 2017; Hairston et al., 1968; Letten & Stouffer, 2019; Pomerantz, 1981). Most past definitions of HOIs based on formal mathematical criteria are problematic because they lump different ecological phenomena together under the umbrella of a HOI or interaction modification. Non-linear density dependence poses particular challenges.

Unlike the Lotka–Volterra example above (Equation 2), many commonly used models of competitive interactions assume a non-linear relationship between per capita population growth and the summed effects of each competitor's density. In models with nonlinear density dependence, the simple partial derivative $\frac{\partial F_i}{\partial n_j}$ may no longer provide such a clear way of defining HOIs. For example, consider the multi-competitor Hassell model (Hassell & Comins, 1976) applied to annual plant assemblages, with nonlinear density dependence (per capita growth is a declining, asymptotic function of the summed effect of competitor density):

$$\frac{n_i(t+1)}{n_i(t)} = F_i\left(n_1, \ldots n_m\right) = \frac{\lambda_i}{\left(1+\sum_{j=1}^{m}\alpha_{ij}n_j\right)^{\tau_i}} \quad (5)$$

Here λ_i is the maximum per capita seed production, α_{ij} is the per capita effect of species j on species i and $\tau_i > 0$ allows each focal species to respond differently to the sum of competitor effects. This function has the partial derivative with respect to n_j:

$$\frac{\partial F_i}{\partial n_j} = -\frac{\tau_i\alpha_{ij}\lambda_i}{\left(1+\sum_{j=1}^{m}\alpha_{ij}n_j\right)^{\tau_i+1}}. \quad (6)$$

Thus, even without any obvious HOI terms, nor any terms that are not fully resolvable in pairwise competition, the effect of competitor j on the focal species i is a function of the density of all other competitor species. This follows from the fact that competition is incorporated into the denominator of a the Hassell model, making growth a nonlinear function of the summed effects of all competitors. Given this, all competitors in the system determine where a focal individual lies on its growth–density relationship, and therefore the effect of any other individual added to the system (Figure 2a). These same points can also be seen in an alternative mathematical context in Supporting Information Appendix A, where we Taylor expand Equation 5, and reveal higher-order terms that are simple functions of the pairwise interaction coefficients. Also in Appendix A, we show another case where all species' densities enter into the partial derivative definition for a HOI, but this time for a continuous time, mechanistically mediated model. Note that in contrast to cases where growth is a nonlinear function of the summed effect of all competitors, the partial derivative definition will not indicate HOIs when growth is linearly related to the sum of nonlinear functions of the density of single competitors (i.e. raising n_j in Equation 2 to a power x). The main message is that the partial derivative definition may not distinguish non-linear density dependence from those interactions only emerging in systems with more than two competitors. This message has been appreciated for some time (Adler & Morris, 1994; Pomerantz, 1981), and Adler and Morris (1994) present a method of substituting the growth expression F_i into the partial derivative to separate off the effects of nonlinear density dependence (Appendix A).

Should we treat nonlinear density dependence as a HOI? Following Abrams (1983) and Adler and Morris (1994), the answer likely depends on the (unseen) mechanistic process driving the nonlinear density dependence. Consider a case where each individual of species j or k consumes a fixed amount of soil nitrogen irrespective of other individuals in the system – setting up a case where there should be no HOIs (Figure 2b). However, if the per capita growth rate of species i is a positive concave up function of nitrogen availability, the reduction in nitrogen due to first competitor individual will have a greater

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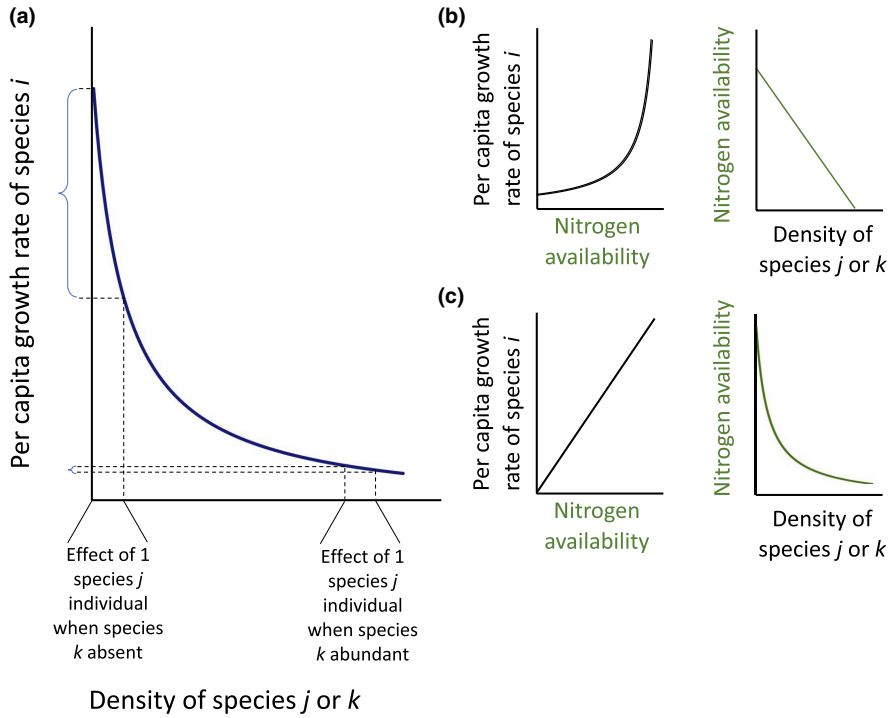


FIGURE 2 A. Due to a nonlinear response of species *i* 's per capita growth rate to the density of its competitors, one species *j* individual has a much greater effect on species *i* when species *k* is absent versus when *k* is abundant. For simplicity, species *j* and *k* are shown exerting identical effects on species *i*, although the same result emerges when their (nonlinear) effects differ. The nonlinear relationship in panel A could emerge from B. a case where each individual of species *j* and *k* identically reduce the availability of a limiting nitrogen resource, but species *i's* per capita growth response to nitrogen is nonlinear. The relationship in panel A could also emerge from C. a case where species *j* and *k* reduce the availability of a limiting nitrogen resource in a density-dependent manner, while species *i's* per capita growth response to nitrogen is linear

effect on the growth of species *i* (by reducing nitrogen where it matters most for growth) than the 10th competitor individual. As a consequence, and despite the fact that species *j* and *k* have no effect on one another's nitrogen consumption, the effect of a species *j* individual on species *i* will depend on the density of species *k*, since the latter determines where species *i* falls on its growth–nitrogen curve. This mechanistic scenario for nonlinear density dependence would presumably not often be described as interaction modification.

Now consider the case (Figure 2c) where species *i's* growth is a *linear* function of resource availability, and the 10th competitor individual exerts less effect than the first because all competitor individuals are smaller and therefore use less nitrogen in crowded conditions. Again, the effect of a species *j* individual on species *i* will depend on the density of species *k*, but in this case, one could argue that species *j* individuals are modifying species *k* individuals (by making them smaller) in a way that determines their effect on species *i*. This scenario for nonlinear density dependence is more easily regarded as interaction modification.

Thus without a known mechanism, it is challenging to interpret the partial derivative of the growth function with respect to a competitor's density as interaction modification or not, and critically, the mechanism underlying the nonlinear density dependence is unseen in the phenomenological model. Moreover, if we accept the partial derivative definition for HOIs as sufficient, we have to conclude that nearly every interaction has a HOI (of this kind) because non-linear density dependence is the predominant form fit to plant competition (Harper, 1977). And of course, if just about every fitted competition model includes a HOI, then the utility of the term is lost (see Adler & Morris, 1994 for a similar argument). It might, therefore, be useful to have different terminology for HOIs defined by interaction modification in the most general sense, and those interactions that uniquely complicate understanding *m*-species dynamics from pairwise processes.

A proposed distinction: soft higher-order interactions versus hard higher-order interactions

To resolve this issue, we propose a distinction between two forms of HOI: (1) 'soft HOIs' which broadly include all interaction modifications, including those resolvable in pairwise trials and possibly manifested in non-linear density dependence and (2) 'hard HOIs' – those interaction modifications only observable in systems with more than two competitor species. In keeping with the literature on HOIs (Billick & Case, 1994), soft and hard HOIs in our structure of definitions can only be identified in

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a mathematical model. They can nonetheless describe processes in nature when such models have been fit to field data. By 'model' we mean an ordinary differential equation or its discrete time equivalent that describes population growth in relation to competitor densities, and competitor densities are the only state variables in the system.

As explained above, soft HOIs are implicit in any model with non-linear density dependence, and their presence can be assessed by taking the partial derivative of the interaction model with respect to the density of any species j (see examples in Table 1). If this partial derivative includes terms for the density of any species other than j (including the species that is the recipient of competition) we can infer soft HOIs. Soft HOIs have an expansive meaning and they are straightforward to test for, but may not suit the needs of those aiming to identify the interaction modifications that only emerge in systems with more than pairwise competition.

Although every hard HOI is also a soft HOI, hard HOIs are specifically those interaction modifications that result in multispecies community dynamics not predictable from pairwise interactions, regardless of the linearity of the relationship between growth and competitor density (see examples in Table 1). Translating this verbal understanding into unambiguous mathematical criteria has in the past led to complicated definitions that never prove sufficiently general (Adler & Morris, 1994; Billick & Case, 1994; Case & Bender, 1981, Supplementary Information Appendix A). Instead we propose a simple empirical heuristic for determining whether a model (fit to empirical data) has a hard HOI. We ask: *Does the model (or family of models) include interaction strengths or interaction modifications an empiricist cannot observe when the competitor species are grown only in pairwise trials* (Pomerantz, 1981)? A hard HOI is a parameter that appears in the multispecies model but is absent in all possible two species versions of that model. Note that this heuristic follows from the central motivation to study HOIs in the first place – we want to know whether a perfect understanding of pairwise interactions between competitors fails to predict the dynamics of the system when all species compete simultaneously. Following the above heuristic, to parameterise a model with hard HOIs, the response of the focal species must be measured against varying combinations of multiple competitors. Hard HOIs are simply those interaction strengths one cannot observe in pairwise competition. See Supporting Information Figure S1 for a graphical interpretation of the heuristic for hard HOIs we propose here. Appendix A presents a possible direction for mathematical generalisation of our hard HOI heuristic based on matrix and tensor rank.

TABLE 1 Examples of models* with no higher-order interaction (HOI), a soft HOI or a hard HOI

No HOI: For all competitors j, $\frac{\partial F_i}{\partial n_j}$ is a function of the density of only species j

$$F_i = \frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \alpha_{ii}n_i - \alpha_{ij}n_j - \alpha_{ik}n_k\right)$$

$$F_i = \frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \alpha_{ii}n_i - \alpha_{ij}n_j - \alpha_{ik}n_k - \beta_{i(ii)}n_i^2 - \beta_{i(jj)}n_j^2 - \beta_{i(kk)}n_k^2\right)$$

Soft HOI: For at least one competitor j, $\frac{\partial F_i}{\partial n_j}$ is a function of the density of a species other than j

$$F_i = \frac{n_{i,t+1}}{n_{i,t}} = \frac{\lambda_i}{1 + \alpha_{ii}n_i + \alpha_{ij}n_j + \alpha_{ik}n_k}$$

$$F_i = \frac{n_{i,t+1}}{n_{i,t}} = \frac{\lambda_i}{(1 + \alpha_{ii}n_i)(1 + \alpha_{ij}n_j)(1 + \alpha_{ik}n_k)}$$

$$F_i = \frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \alpha_{ii}n_i - \alpha_{ij}n_j - \alpha_{ik}n_k - \beta_{i(ij)}n_in_j\right)$$

Hard HOI: For at least one competitor j, $\frac{\partial F_i}{\partial n_j}$ is a function of the density of a species other than j, AND some model parameters are absent from the set of all two species versions of the model (species i and j model where $n_k = 0$ and species i and k model, where $n_j = 0$)

$$F_i = \frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \alpha_{ii}n_i - \alpha_{ij}n_j - \alpha_{ik}n_k - \beta_{i(jk)}n_jn_k\right)$$

$$F_i = \frac{1}{n_i}\frac{dn_i}{dt} = r_i\left(1 - \alpha_{ii}n_i - \alpha_{ij}n_j - \alpha_{ik}n_k - \beta_{i(jk)}n_j^2n_k\right)$$

$$F_i = \frac{n_{i,t+1}}{n_{i,t}} = \frac{\lambda_i}{1 + \alpha_{ii}n_i + \alpha_{ij}n_j + \alpha_{ik}n_k + \beta_{i(jk)}n_jn_k}$$

*n_i is the density of species i, r_i and λ_i are its instantaneous and multiplicative intrinsic growth rate, respectively, and α_{ij} is the per capita effect of species j on the growth of species i. $\beta_{i(jk)}$ describes the interactive effect of species j and k on the growth rate of species i. F_i is the per capita growth rate of species i.

What complicates matters is that even in a world with no hard HOIs, a perfect understanding of pairwise interactions does not reveal the m-species dynamics. This is because regardless of HOIs, a set of well-parameterised pairwise interaction models can be combined in multiple ways to generate an m-species model. And thus parameterising models with more than two species always requires focal individuals simultaneously confronting multiple competitor species at different densities, and this true is regardless of one's interest in testing for soft or hard HOIs (Adler & Morris, 1994).

For example, consider the pairwise effects of two competitors, j and k on species i:

$$\frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(1 + \alpha_{ij}n_j\right)^{\tau_i}} \quad \text{and} \quad \frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(1 + \alpha_{ik}n_k\right)^{\tau_i}}.$$

There are multiple forms of multispecies competition models with no hard HOIs that collapse to these pairwise models (by setting one of the competitor's densities to zero), including:

$$\frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(1 + \alpha_{ij}n_j + \alpha_{ik}n_k\right)^{\tau_i}} \quad \text{and} \quad \frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(1 + \alpha_{ij}n_j\right)^{\tau_i}\left(1 + \alpha_{ik}n_k\right)^{\tau_i}}$$

Only by growing individuals of species i with varying densities of species j and k, can one determine which multispecies model fits best. Although this question of multispecies model structure can never be answered with just an understanding of the pairwise interactions, this lack of resolution is simply a property of the model fitting process, not itself a HOI. Once a best fit model is

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decided upon, one can then ask if it includes soft and hard HOIs.

For the models in the prior paragraph, we would deduce that both have soft HOIs due to nonlinear relationships between growth and competition. But neither have hard HOIs because in both, the per capita competitive effects α_{ij} and α_{ik}, (and τ_i) can be fit by growing individuals of species i in pairwise competition separately with j and k.

Our proposition to separate hard HOIs from nonlinear density dependence follows from Adler and Morris (1994). In short, their quantitative approach modifies the partial derivative definition for HOIs (for what we call 'soft HOIs') to account for nonlinear density dependence (Supporting Information Appendix A). Their criteria for 'interaction modification' generates a similar categorisation of models as our heuristic for hard HOIs. We have, however, found that in diverse systems with multiple forms of density dependence, the Adler & Morris approach can indicate HOIs when all the interaction strengths are measurable in pairwise competition (Supporting Information Appendix A), reflecting the different motivations of two approaches. We thus believe our heuristic is potentially more flexible if the goal is specifically to identify interactions only emerging in systems with three or more species.

Why distinguish hard and soft higher-order interactions?

From the standpoint of predicting coexistence or community change, accounting for both soft HOIs and hard HOIs can increase the accuracy of model fit and predictions (Letten & Stouffer, 2019). Moreover, as described earlier, both classes of HOI can be interpreted as interaction modification, albeit ambiguously so in the soft HOI case. However, only hard HOIs represent a fundamental challenge to our existing pairwise theories of ecology. In contrast, the question of whether the decline in population growth rate with competitor density is linear or non-linear is a longstanding issue in ecology (Hassell & Comins, 1976), and it is not clear that further work seeking out soft HOIs will generate new conceptual insights.

Our definition also bears on whether single species effects can involve HOIs. For instance, recent papers by Letten and Stouffer (2019) and Mayfield and Stouffer (2017) define HOIs as any higher-order terms of competitor density, including single species quadratic terms, $\beta_{i(jj)}n_j^2$ (i.e. second equation of Table 1). Our definition does not count these as soft or hard HOIs, as the effect of species j on i is affected by no species other than j, and these interactions are entirely resolvable in pairwise competition. This follows the tradition in the literature where HOIs are a phenomenon that arise due to the interactive effects of *different species* of competitor (Hairston et al., 1968; Morin et al., 1988; Neill, 1974; Vandermeer, 1969). As per our definition, the coefficients for these terms, $\beta_{i(jj)}$, are parameters in a pairwise (or even single species- $\beta_{i(ii)}$) model. The addition of these quadratic terms to the model simply allows a better nonlinear fit to the form of density dependence.

More subtle is the categorisation and interpretation of interaction modifications that operate within two species systems – that is, the addition of a $\beta_{i(ij)}n_in_j$ to the Lotka Volterra or Hassell models (e.g. the third soft HOI model in Table 1). Clearly, these interactions would constitute soft HOIs as species j alters the effect that species i has on itself; if $\beta_{i(ij)}n_in_j$ were added to a Lotka Volterra model, the partial derivative test for soft HOIs would identify one ($\frac{\partial F_i}{\partial n_j}$ is a function of the density of a species other than j). However, following the heuristic for hard HOIs posed here, such 'two species interaction modifications' are not hard HOIs since they are entirely resolvable with a perfect understanding of pairwise competition. In other words, fitting the parameter $\beta_{i(ij)}$ requires experiments with no more than two species. Such interaction modifications are likely to be dynamically important, but they do not uniquely complicate m-species dynamics.

DETECTING AND INTERPRETING HIGHER-ORDER INTERACTIONS IN A SIMULATED COMPETITION EXPERIMENT

Having established a distinction between soft and hard HOIs in mathematical models of competition, how does this structuring of definitions for HOIs influence how we test for these interactions with real data? More specifically, how does one obtain a quantitative description of pairwise competition sufficient to test for soft and hard HOIs, and distinguish between them? In the remainder of this article, we walk through the primary considerations that an ecologist encounters in efforts to detect and interpret soft and hard HOIs with field data.

Because empirical data allowing for rigorous tests of HOIs are limited, we use a mechanistic growth model (Box 1) to simulate a virtual competition experiment among three annual plant species competing for water, and differing in their period of maximal growth (Figure S3). We then fit phenomenological competition models with soft and hard HOIs to species' growth responses to their competitive environment. More than providing a recipe, this exercise demonstrates the fundamental challenges of determining whether hard HOIs operate in a real-world setting.

An experimental design to characterise both soft and hard higher-order interactions

Given that much of the confusion about how to define HOIs follows from nonlinear density dependence,

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BOX 1 THE MECHANISTIC COMPETITION MODEL

The model simulates the growth of annual plants competing for water in a Mediterranean climate (Figure S2). The simulated individuals germinate in the winter and then grow, and produce seeds by the early summer (Godoy & Levine, 2014). In the model, we track a single pool of soil water, supplied only once during the year (winter), and depleted over time by the plants. As water availability declines, plant growth slows and eventually stops (Figure S3). Competition occurs in our simulations because as individuals grow, they draw down water (Figure S3), and this decreases the rate of water acquisition into the roots and slows the growth of other individuals in the community. We make the assumption that when individual net growth is zero, the plant converts a fraction of its biomass into seeds that will survive as dormant individuals and germinate at the start of the next growing season (Cohen, 1976). Per capita production of these seeds is therefore a direct measure of population growth rate.

The plants differ in their phenology due to their differential allocation to roots and, therefore, their ability to uptake water from the soil as it dries through the season. This difference causes species to differ in the soil water availability / time at which they stop growing (Figure S3), and the three species we simulate are therefore referred to as 'early', 'mid' and 'late' (Table S1). A detailed summary of the mechanistic growth and competition model is provided in Supporting Information Appendix B. Although a mechanistic model this simple could be fully informed by field experimental data, obviating the need for fits to phenomenological models, it was chosen to be one of the simplest of models that would generate data we could evaluate for soft and hard HOIs.

The pairwise interactions we fit to the mechanistic model (Figure 6a–c) can be understood from the overlap in growth period between competitors. For instance, the late season species has a weak per capita effect on the early season species because while the early species is active, the late-season species remains small and has a slow absolute rate of water uptake (Figure S3). In contrast, the mid-season species has a stronger effect on the early season species because it grows faster during the same period (Figure S3). On the other hand, the early season species has a weak effect on the late season species because the former stops growing and competing well before the latter does the majority of its growth (Figure S3).

We can also explain the emergence of HOIs in our model from the period of overlap between the competitors, although a detailed mechanistic understanding the HOIs in our simulation model is beyond the scope of this article. The early species accentuates the interactions between the mid and late species by suppressing these taxa early in the year. This early season suppression lengthens the period of time the mid and late taxa are growing and competing, strengthening their interaction (Figure 6e,f). By the same reasoning, the early species experiences no HOIs because over its entire growth period it experiences competition by the mid and late species, and this is unchanged by their abundances (Figure 6d). The two species soft HOIs (the $\beta_{i(ij)}$'s in Figure 6) can be similarly explained. With increasing density of the later species of the pair, that species finishes its growth earlier in the year, increasing its overlap with the earlier competitor and accentuating the effect of that competitor.

an experimental design testing for soft and hard HOIs must allow for two types of inference – (1) the form of density dependence and (2) interaction modification in settings with more than two species. Making both inferences requires measuring how a focal species' performance changes in response to the increasing density of each competitor species in isolation, as well as to varying combinations of densities of the different competitor species. This can be achieved with an orthogonal response surface design where each competitor's density is varied independently of each other species (Figure 3).

We quantified all species' per capita population growth rates in plots with densities of 0, 1, 2, 3, 4, 9, 16, 25 or 36 individuals of each of the competitor species in isolation, with the same densities applied to measure intraspecific competition. We also measured performance when individuals faced all possible combinations of two competitor species (Figure 3). This design allows one to first describe the pairwise interaction effects, nonlinear or otherwise, when only two species compete, and then test for the changes in competition that occur when a third competitor species is added to the mix. Note that a design where focal individuals only face two competitor densities (e.g. a classic two factor ANOVA design), which generates just two points in a relationship between growth and density, cannot resolve the shape of the density dependence, and thus cannot distinguish soft from hard HOIs.

In principle, one could fit a wide range of multispecies models to data from the full response surface in Figure 3 and then evaluate the models for soft and hard HOIs.

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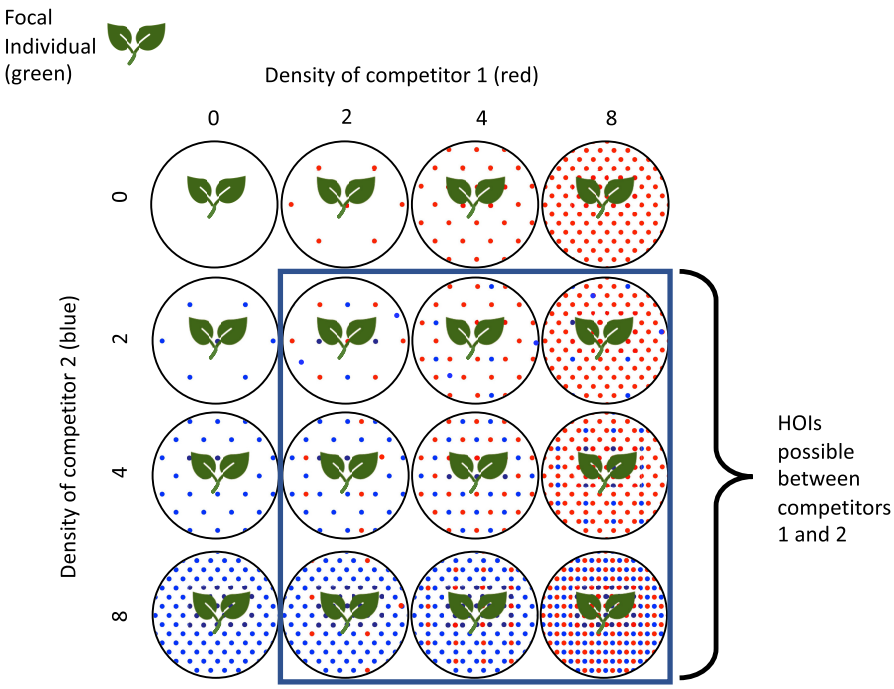


FIGURE 3 Schematic of a competition experiment designed to detect higher-order interactions. Each circle represents a separate plot. Competitor 1 (red) and Competitor 2 (blue) are grown in a gradient of increasing density alone and together. A single individual of the focal species (green) is grown in each plot, allowing the response to competition from each competitor species to be fitted

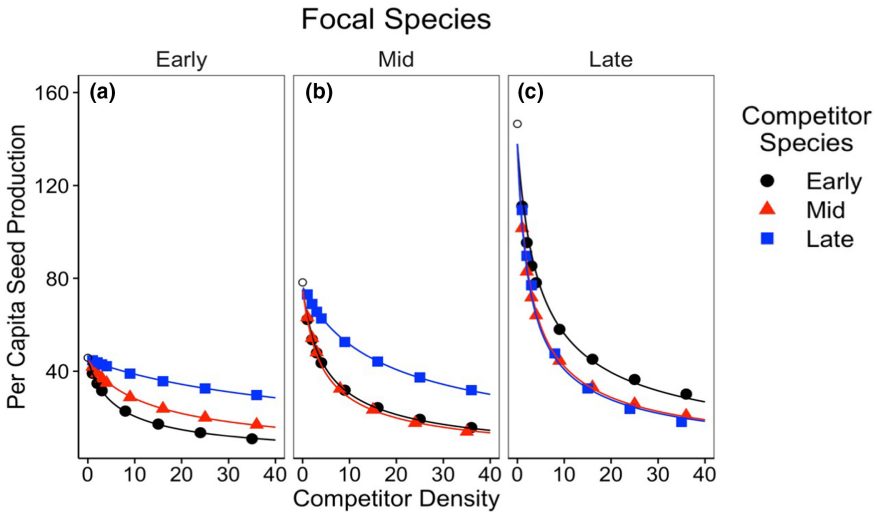


FIGURE 4 Simulated per capita seed production of the (a) early, (b) middle and (c) late season species in response to a single competitor species at a time. Competitor density is shown on the x-axis. Colours and shapes indicate the identity of the competitor species. Open circles show the per capita seed production of each focal species in the absence of any competitors. The solid lines show the fit of the Hassell model

However, if one begins with a range of reasonable pairwise interaction models, examines all reasonable ways those pairwise interactions combine, and then examines all reasonable additions of hard HOI terms to those models, one quickly ends up with more models than can be fit from available data. Therefore, in what follows, we outline a series of practical, sequential steps towards honing in on the best fit model, which can then be tested for soft and hard HOIs.

Quantify the nonlinearity of pairwise competition

If the ultimate goal of HOI research is to explore whether an understanding of pairwise competition adequately predicts the dynamics of all species simultaneously interacting (testing for hard HOIs), a good description of pairwise interactions is required before attempting to describe multispecies systems. In fact, a poor choice of pairwise interaction model will likely benefit from the

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inclusion of higher-order terms because the more flexible forms of density dependence these terms afford will compensate for the poor model choice. This ambiguity may not matter for investigations aimed only at soft HOIs, although testing for hard HOIs (deviations from pairwise predictions) implicitly requires confidence in one's understanding of pairwise effects, a point we come back to in the discussion.

Examining the plots of seed production per individual against the density of each competitor species in isolation (Figure 4) clearly shows that seed production declines non-linearly as the density of inter- or intraspecific competitors increases. This rules out a simple Lotka–Volterra form to the decline in per capita growth rate with competitor density and suggests the operation of soft HOIs. For our simulated experiment, we find that the Hassell model (Equation 5) fits the outcome of simulated pairwise competition well (Figure 4). We fit this and all subsequent models with the non-linear least squares modelling function, `nls`, in R.

Fit the multi-competitor results to multiple models with only soft higher-order interaction terms

To further refine our search for an adequate multispecies model, we need to address how the pairwise interactions combine to describe the fate of individuals facing multiple competitor species. Only once that multispecies model structure combining pairwise interactions is specified does it make sense to consider the effects of hard HOIs. To this end, we first fit the *m*-species Hassell model (with no hard HOIs, Equation 5) to the entire dataset, now including the plots where two different species compete with our focal species (Figure 5). For each focal species and model, we calculated the root-mean-squared error (RMSE) as a measure of goodness of fit. We find that the *m*-species Hassell model does not fit the performance of the mid- and late season species as well as it does for the early species (Figure 5 and Figure S4).

Do the deviations from the Hassell model for plants in multispecies competition indicate that hard HOIs are operating? Before answering, we need to ensure that the multispecies model built from the pairwise components of the Hassell model is the best amalgamation of our pairwise information – that is, that we have a reasonable multispecies model structure. As introduced at the end of the earlier section *A proposed distinction*: *soft higher-order interactions versus hard higher-order interactions*, depending on the best-fit pairwise models, several multispecies models that collapse to these same pairwise fits can be considered. And there is nothing in principle superior about the way the *m*-species Hassell model additively combines species' pairwise effects, versus for example, the following multiplicative alternative:

$$\frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(\prod_{j=1}^{3}\left(1+\alpha_{ij}n_j\right)\right)^{\tau_i}}. \quad (7)$$

Note that this model includes soft HOIs based on the partial derivative definition. But following our heuristic, this model does not include hard HOIs because the α_{ij}'s (and λ_i and τ_i values) can be estimated when the focal species i competes with all other species j in isolation of one another.

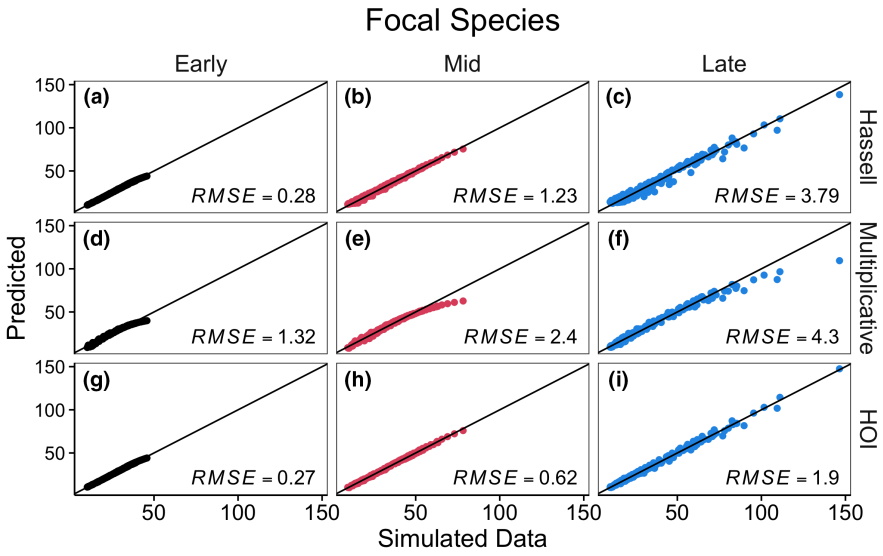


FIGURE 5 Comparison of three phenomenological models (Hassell, Multiplicative and higher-order interaction [HOI] – each described in the main text) fit to each focal species. The x-axis shows the simulated per capita seed production of the focal species. The y-axis shows the per capita seed production predicted by the phenomenological model. The top row, (a)–(c), shows the prediction for the pairwise Hassell model; the middle row, (d)–(f), shows the prediction from the multiplicative model; and the bottom row, (g)–(i), shows the prediction from the HOI model. The one-to-one line and root-mean-squared error (RMSE) for predictions from each model are shown

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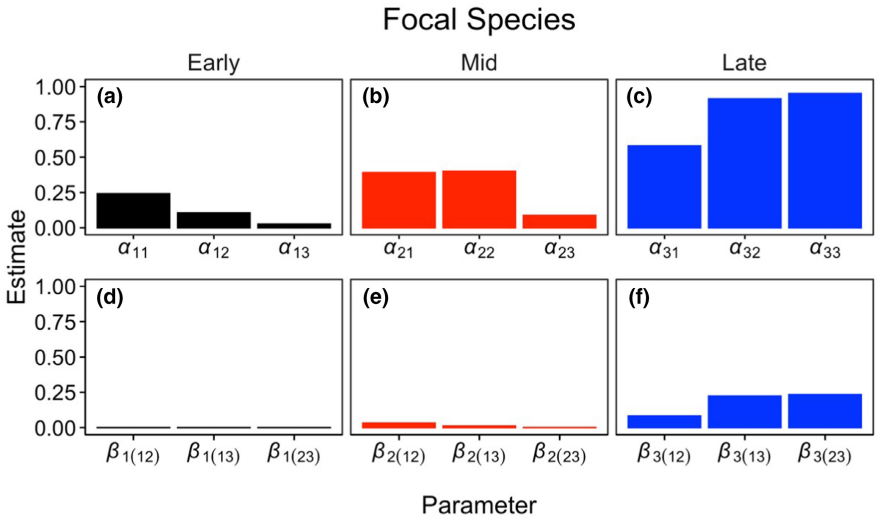


FIGURE 6 Interaction coefficients for each of focal species from the HOI model. The top row, (a)–(c), shows the pairwise competition coefficients for the focal species and each competitor. The bottom row, (d)–(f), shows the HOI coefficients. Coefficient subscripts indicate which focal species and competitor species are involved, 1 = Early, 2 = Mid, 3 = Late

When we fit this multiplicative model (Equation 7) to the multi-competitor experimental results, it did poorly compared with the multispecies Hassell model (Figure 5; Figure S4). Thus, we conclude that however poor the fit of the Hassell model to the mid and late species, the fit is not improved by this alternative multispecies model structure. One could fit further combinations of the pairwise Hassell model, but we see no straight-forward alternatives.

Fit the multi-competitor results to models with hard higher-order interaction terms

Once we have narrowed down a multispecies model structure, we can include a three species hard HOI and two species soft HOIs in the three species Hassell model as follows:

$$\frac{n_i(t+1)}{n_i(t)} = \frac{\lambda_i}{\left(1+\sum_{j=1}^{3}\alpha_{ij}n_j+\beta_{i(ij)}n_in_j+\beta_{i(ik)}n_in_k+\beta_{i(jk)}n_jn_k\right)^{\tau_i}}, \quad (8)$$

where all HOI effects of two competitor species on the focal species i are fitted with the coefficients $\beta_{i(jk)}$ (following the notation in Mayfield and Stouffer (2017)). By our definition, the parameter $\beta_{i(jk)}$ captures a potential hard HOI because it can only be fitted in settings where two or more competitor species (j and k) are present together. Other more complex forms of hard HOI – which disappear in pairwise contexts – are also possible (e.g. $\beta_{i(jk)}n_j^2n_k$ in Table 1). But a simple HOI that scales the product of species densities as in Equation 8 is the first step in such an analysis.

For the early season species, the Hassell model with and without the two and three species HOIs showed more or less equivalent fits to the data with only a slight decrease in RMSE for the HOI model (Figure 5a and g). For the mid-season and late-season species, we found that the HOI model fit the data better than the pairwise Hassell model (Figure 5b,c,h,i). The superior fit of the HOI model (with $\beta'_{i(jk)}s$) relative to pairwise Hassell model (no $\beta'_{i(jk)}s$) for the mid and late-season species can also be seen by plotting the observed per capita seed production against two competitor densities at once (Figure S4). In all cases, the fitted HOI coefficients, $\beta_{i(jk)}$, were of smaller magnitude than the fitted pairwise effects, α_{ij} (Figure 6), but this is to be expected given that α_{ij} is a scalar on density, while $\beta_{i(jk)}$ is a scalar on the product of densities. The fitted HOIs were stronger for the mid and late season species than for the early season species (Figure 6), and the two species soft HOIs ($\beta'_{i(ij)}s$) were of greater magnitude than the three species hard HOIs ($\beta'_{i(jk)}s$). This latter result likely follows from the simple fact that the pairwise interactions modified by the soft HOIs were of greater magnitude than those modified by the hard HOIs (Figure 6). Together this evidence supports the interpretation that hard HOIs operate in this system of three annual plant competitors. As outlined in Box 1, this follows from the way the early species lengthens the duration of competition between the mid and late species.

DISCUSSION

HOIs have profound implications for how we understand and model multispecies communities. However, before ecologists can properly embark on measuring HOIs in nature, we must sort out what HOIs are. Our distinction between soft and hard HOIs presents a nested structure of definitions that follows from the two primary motivations for this research effort. Soft HOIs, which identify possible interaction modification, can be identified by calculating a partial derivative of one species growth function with

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respect to the density of another. Hard HOIs, which identify interactions uniquely emerging in systems with three or more competitors, can be identified by asking if any interactions in an *m*-species model are unseen in pairwise competition. By simulating an experiment with results driven by a mechanistic model of competition, we have shown that these definitions can be operationalised in experimental settings. In the course of doing so, however, we have identified a number of ambiguities and potential arbitrariness in this research effort that we elaborate on in this discussion. We conclude by arguing that despite these challenges, further developing this research area remains an important endeavour for understanding the dynamics of species in diverse communities.

Better framing the motivations for studying higher-order interactions

The fundamental challenge when studying HOIs is that the two seemingly straightforward (and related) motivations for research are imprecisely framed. On the one hand, research on HOIs is motivated by an interest in how the effects of one species on the interactions between another two (or more) changes the dynamics of the system as a whole. However, in contrast to interaction modification in a trophic context, where it tends to be mechanistically motivated (Schmitz et al., 1997), exactly what constitutes interaction modification in competitive systems is unclear. For example, Adler and Morris (1994) see nonlinear density dependence as something separate from interaction modification, we see it as ambiguous, whereas other efforts include these effects as HOIs (Mayfield & Stouffer, 2017). Finally, as noted earlier, simply a poor choice of pairwise model might underlie the significance of higher-order terms that would conventionally be regarded as interaction modifications. Given these interpretation challenges, we call HOIs based on this interaction modification motivation soft HOIs – they may or may not involve interaction modification. Regardless, there exists a clear partial derivative test that unambiguously identifies such HOIs in models fit to data from nature (Table 1).

The second motivation to study HOIs is the desire to understand if one can predict the dynamics of a species rich system from its component pairwise interactions (Billick & Case, 1994). The challenge with this framing is that regardless of the operation of HOIs, one can never know the *m*-species dynamic with only knowledge of the pairwise dynamics. In other words, even without hard HOIs there are inevitably multiple ways to combine the pairwise components, as we showed with the Hassell model fit to our *in silico* data – what we call the question of 'multispecies model structure'. Thus, the more precise way to phrase the second motivation for studying HOIs, as is sometimes done in the literature, is to reverse the question. Does a multispecies model describing nature include interactions that cannot be observed in pairwise competition? Our heuristic for identifying hard HOIs is exactly that.

Interpreting hard higher-order interactions

Our definition of hard HOIs provides a practical way forward for research aiming to understand if some processes uniquely emerge in multispecies systems. Still, as we noted earlier in the article, the search for the best fit multispecies model does not have an objective stopping point, and even the correct functional form for the HOI term is not a given. Although restricting our proposed structure of definitions to models that ecologists commonly fit to data may lesson these concerns, doing so also raises the fundamental question of how to interpret hard HOIs.

It is tempting to treat statistical support for models with hard HOIs as evidence that species interactions in systems with more than two competitors involve some fundamentally different process than operates in pairwise cases. And this certainly may be the case for some hard HOIs. For these interactions, even if one had a mechanistic rather than phenomenological description (model) of the underlying process driving the interactions, some parameters could only be described by growing multiple competitors together (i.e. the plastic response of one competitor's morphology cued by another competitor's density). Similarly, in a trophic context, quantifying how an herbivore's behavioural response to a predator regulates its impact on plants may require growing all species together (Schmitz et al., 1997).

However, hard HOIs only reveal that for the phenomenological models fit to data, key parameters are absent from all pairwise subsets of those models. Thus, the inference is only as good as the pool of models being fit, and as we have already noted, when the fitted models are clearly poor choices, the hard HOIs are hardly diagnostic for emergent properties in multispecies systems. One must be particularly mindful about the time scale over which the model advances. For example, Grilli et al. (2017) identify HOIs in a theoretical model where all competition occurs over short-time scales among tree seedlings, and does so in a clearly pairwise manner. But over the longer timescale necessary to understand the coexistence of adult trees, repeated (within time interval) interactions between seedlings generate the HOIs. Similarly, had our own phenomenological models defined per capita effects of competition on plant biomass, advancing on a daily rather than annual time scale, and explicitly accounted for the change in the duration of growth for each species in different competitive settings, our data may have been well-fit by models without HOIs. Thus the interpretation of HOIs supported by models fit to

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data is always conditioned on the time scale over which dynamics proceed in the fitted model.

But the problem is even more complex when considering how data are brought to bear on the problem. Even in cases where the pairwise fits to data appear to be excellent, and a hard HOI is supported by the multispecies data, it is always conceivable that a more complex model with all parameters observable in pairwise trials fits the data as well or better than the HOI model. But one never knows that model, and to make matters worse, data collected from real competition experiments will be vastly noisier than the simulated data in our hypothetical experiment, making it much harder to confidently conclude that the "correct" model has been fitted. We do believe that the sequential fitting process we have employed here is a more pragmatic approach than simultaneously confronting data with the wide space of possible pairwise and HOI models. Although even if one takes this approach, one can never distinguish between what might be regarded as "true" hard HOIs and hard HOIs that are a function of the practical limitations of how we fit models to data.

It might, therefore, be most productive to simply treat hard HOIs as evidence that one is modelling or conceptualising the system in too simple a manner, and bring in other evidence to distinguish the different interpretations of hard HOIs. This could include measurements of how the functional traits determining competition change in the presence of different competitors, or how the apparent HOIs vary with environmental factors or the presence of other trophic levels.

Why not go mechanistic?

One possible solution to the challenges inherent in testing for HOIs is to sidestep the issue entirely and investigate only mechanistic models of community dynamics. For the model in Box 1, for example, one could envision fitting the resource uptake functions, biomass to seed conversion rates, and other parameters. Might this approach more productively identify the processes controlling system dynamics than fitting phenomenological models with pairwise and higher-order terms? Mechanistic models do not suffer the challenges of HOIs as the multispecies dynamics simply follow from the functions relating competitors to their resources and/or consumers. Moreover, the results of such modelling efforts give a process-level understanding of the controls over system dynamics, extrapolatable to situations in which the driving variables change. Thus, for some research aims, mechanistic models are clearly the way forward.

In contrast to mechanistic approaches, phenomenological models have the advantage of being able to accommodate many different possible mechanisms of competition (Chesson, 2000), and their fit to data does not require a prior knowledge of the mechanisms of the interaction. HOIs emerge in such models precisely because they leave out mechanistic detail and do not explicitly model the dynamics of resources, consumers and pathogens (Abrams, 1983; Letten & Stouffer, 2019; O'Dwyer, 2018), a feature motivating many to model communities mechanistically (e.g. Dybzinski and Tilman (2007)).

However, from a pragmatic perspective, few ecological systems are simple enough to be well described by mechanistic models; a notable exception being trees competing for light (Strigul et al., 2008). And mechanistic models do not allow species interactions to be compared and described using a simple set of parameters. This means that mechanistic models cannot, in and of themselves, address the two primary motivations for studying HOIs – interaction modification and the emergence of process in diverse systems. In fact, a recent effort to use mechanistic models to this end involves an intermediate step of fitting a phenomenological model to the dynamics produced by a mechanistic consumer resource model before HOIs are assessed (Letten & Stouffer, 2019). Meanwhile, efforts to analytically derive the phenomenological interactions from mechanistic model parameters, which is possible in some cases, is typically hampered by the complexities of nature (O'Dwyer, 2018). Thus, even in cases in which we actually know the resources for which species compete, fitting a phenomenological model to population dynamics may be the only way to quantitatively describe species interactions. Finally, from a theoretical perspective, many important questions in ecology, such as how coexistence is maintained, are rooted in a phenomenological conception of species interactions (Chesson, 2000). Therefore, the problem of phenomenological HOIs cannot simply be side-stepped.

CONCLUSION

Over the last half century, ecologists' desire to move beyond the well-acknowledged limitations of building a theory of community ecology through pairwise interactions has repeatedly energised interest in HOIs. Each time, however, these bouts of energy have fizzled due to the logistical challenges of obtaining the necessary data as well as shifting definitions, which have made the empirical work and statistical tools sufficient to detect HOIs both a moving target and increasingly out of reach. Here, we have proposed a practical and justifiable structure of definitions for HOIs, and have shown how this structure can be implemented in an experimental context. We hope this effort to clarify the aims and challenges catalyses further work in this important area of research.

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AUTHOR CONTRIBUTION

All authors conceived of the ideas and approach of this manuscript. ARK developed and simulated the model, analysed the output, and wrote the first draft of the manuscript. All authors contributed substantially to revisions.

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DATA AVAILABILITY STATEMENT

No new data were used in this article; all code is available at https://doi.org/10.5281/zenodo.6466391.

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